LESSON PLAN · SCRATCH (GR 3–5) BEGINNER: SCRATCH GAME MAKERS

Lesson 12: Capstone: My Arcade Game

Module 4 | 90 minutes | Capstone: everything together | Grades 3–5 · ages 8–10

Objective: Design, build, playtest, improve and present an original game that uses events, loops, conditions, variables and messages.

Success criteria

- My game uses events, loops, conditions, a variable and an ending.
- I playtested and improved my game.
- I can explain one clever part of my code.

Key vocabulary

- **capstone**: A final project that uses everything you learned.
- **game design**: Planning how a game works and feels.
- **iteration**: Improving something step by step.
- **playtest**: Testing a game by watching someone play.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (90 min)

Time	Stage	What happens
10 min	**Session 1 · Plan**	Paper prototype; game design plan in Kitabash; teacher checks plans.
30 min	**Session 1 · Build**	Player controls, then the main challenge. Grand-challenge puzzle for early finishers.
5 min	**Session 1 · Save**	Save progress (the Code Lab saves on the device; submit a draft if changing devices).
20 min	**Session 2 · Score & endings**	Variable and win/lose ending.
10 min	**Session 2 · Playtest**	Swap; two stars and a wish; improve one thing.
15 min	**Session 2 · Submit & showcase**	Reflection and submit; 1-minute presentations.

Unplugged warm-up

Paper prototype (10 min): students draw their game screen on paper and “play” it with a partner using paper cut-outs, explaining the rules aloud.

Teaching the idea

The capstone integrates every concept. Students follow a design process (plan → build in increments → playtest → improve → present). Assess with the 6-criterion rubric; the auto-checks confirm the concepts are present, the teacher judges quality and creativity.

Analogy: Building a game is like building a house: plan, foundations (player controls), walls (rules), roof (score and endings), then decorate (polish).

Live demo: Show three short example games (catch, maze, dodge) made in the Code Lab. Model filling in the planner for one of them.

Questions to ask

Ask	Listen for
What’s the goal of your game?	A clear, one-sentence goal.
Which concept are you using for that?	Name the concept (event, loop, condition, variable, broadcast).
What did your playtester struggle with?	Specific observation.

Differentiation

Support
Offer the catch-game template (from Lesson 9) as a starting point; the student personalises sprites and adds a score and ending.

Stretch
Levels (a “level” variable that changes speed), a title screen, or a high score.

Common misconceptions

Misconception	How to fix it
Build everything at once, test at the end	Build and test in small steps.
A game needs lots of sprites to be good	A simple game that works well and is fun is better.
Feedback means my game is bad	Feedback helps every designer improve.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Events & controls	Hard to control	Basic controls work	Smooth, clear controls	Controls feel great; extra events used well
Loops & conditions	Few rules work	Some rules work	Loops and if rules make a working game	Clever combinations (e.g. if-else, nested loops, speed-up)
Variables & endings	No variable or ending	Variable changes but no ending	Variable and a working win/lose ending	Several variables or game states (title, play, end)
Design & creativity	Copy of an example	Some personal choices	Original and fun	Polished, original, balanced and surprising
Process (plan, test, improve)	No plan or testing	Plan or test, not both	Planned, tested and improved	Several improvements from testing, well documented
Reflection & presentation	Missing	Brief	Explains a bug fix and a change	Explains learning clearly and presents confidently

Exit ticket
One thing I'm proud of in my game, and one thing I learned in this course.

Homework
Share your game with family. Ask them to beat your score!

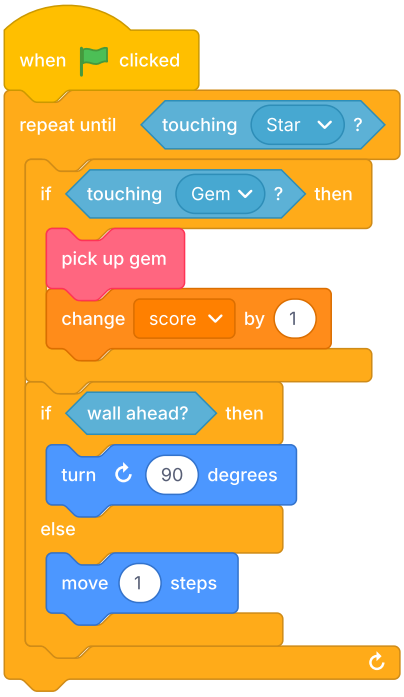
Answer key — Kitabash activities

1. Which big idea does each game feature use? (perform · categorize)

Events: Arrow keys move the player, Clicking start begins the game
Loops: The enemy walks back and forth forever, Music plays all game
Conditions: If the player touches lava, go back, If score = 10, you win
Variables: The score goes up by 1, Lives start at 3

2. Grand challenge (perform · block code)

One solution:



3. Game design plan (probe · free response)

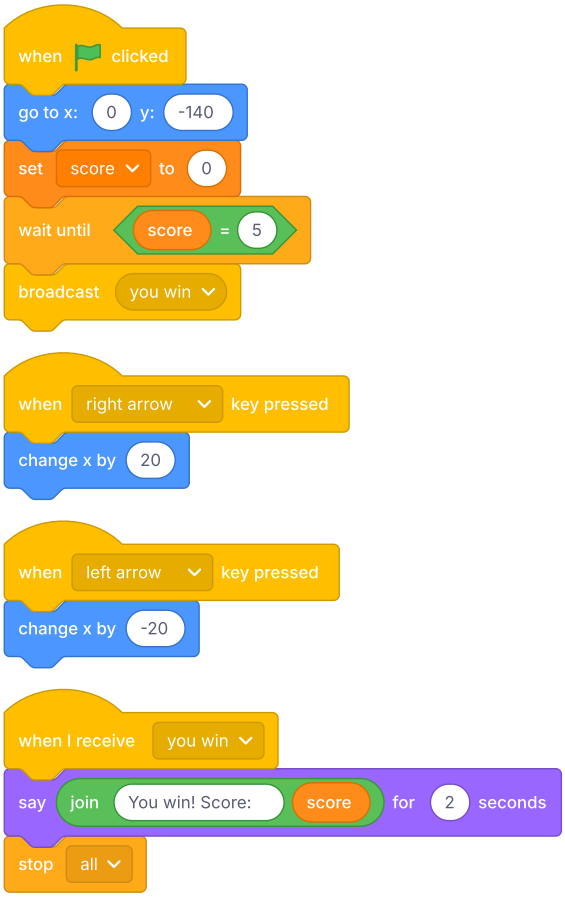
Look for: goal, win, lose, arrow. **Example:** My game is called Space Dodge. The goal is to survive for 30 seconds. The player is a rocket that moves with the left and right arrows. Meteors fall from random places. If a meteor touches me, I lose a life. I win if the timer reaches 30 and lose if lives reaches 0.

4. Capstone: My Arcade Game (probe · code lab)

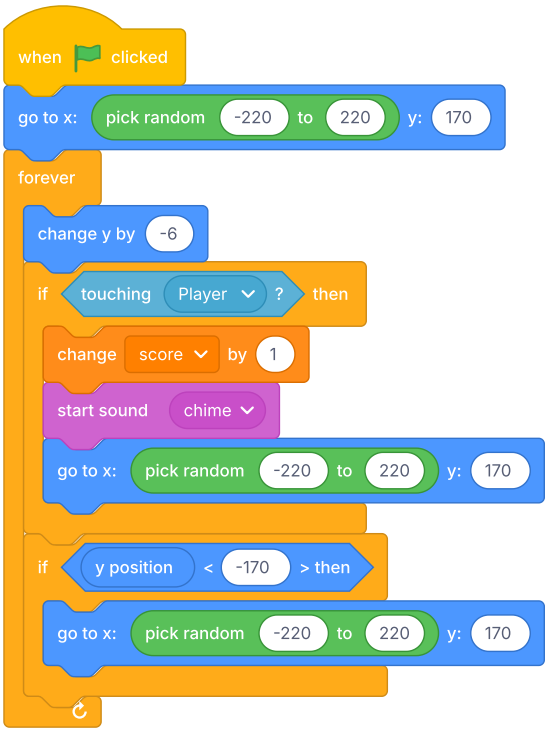
Goals checked: Player controls (keys or clicks) · Uses a loop · Uses an if rule · Uses a variable · Has a win or lose ending · 2 or more sprites

Example project:

Player



Star



Answer key — worksheet

1. **My game:** A named game with a clear goal.
2. **Sketch the screen:** A labelled sketch.
3. **Big ideas checklist:** One specific use for each.
4. **Playtest notes:** Feedback and a change.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.